De la Nube a la FPGA: Introducción Práctica a RISC-V con Quintauris

Enrique Pallares
Head of Information Office

QUINTAURIS

Agenda

2 Semanas – 8 sesiones

SEMANA 1

LUN 7 17:00–19:00
1. Fundamentos de RISC-V: de la arquitectura al ecosistema
Quintauris · Ángel Berrio

MAR 8 17:00–19:00
2. Presentación de Quintauris: visión, modelo de negocio y productos
Quintauris · Ángel Berrio

MIÉ 9 17:00–19:00
3. Explorando la plataforma Quintauris, Tools and Infra — incluyendo Testing, Benchmarking y QNTAF (I)
Quintauris · Enrique Pallares

JUE 10 17:00–19:00
4. Explorando la plataforma Quintauris, Tools and Infra — incluyendo Testing, Benchmarking y QNTAF (II)
Quintauris · Enrique Pallares

SEMANA 2

LUN 14 16:30–19:30
5. IP hardware RISC-V open source: introducción, licencias y repositorios
Quintauris · Ignacio Ruiz

MAR 15 16:30–19:30
6. Parte I: De la nube a la FPGA — flujo de despliegue de una IP en la FPGA
UGR & Quintauris · Jose Luis Martínez, Ignacio Ruiz / UGR

MIÉ 16 16:30–19:30
7. Parte II: De la nube a la FPGA — flujo de despliegue de una IP en la FPGA
UGR & Quintauris · Jose Luis Martínez, Ignacio Ruiz / UGR

JUE 17 16:00–20:00
8. Hackathon final: integración de conceptos y solución de un reto práctico
Quintauris · Jose Luis Martínez, con mentores de UGR

The Quintauris Team

Ángel Berrio

Chief Product Officer

Sessions 1 and 2

Enrique Pallares

Head of Professional Services

Sessions 3 and 4

Ignacio Ruiz

Testing & Validation Staff Engineer

Session 5 and hackathon

Jose Luis Martinez

Application Engineer

Session 5 and hackathon

Session 3

Exploración de la plataforma Quintauris: entorno, funcionalidades, Infraestructura y Framework de Automatización de Test and Benchmarking I

Agenda

01 QNT Specifications

02 Infrastructure

03 Test Automation Framework

04 OS Integrations

05 Wrap-Up & Discussion

The system Problem

Same Instruction Set. Three Different Systems.

WHERE IT BREAKS

- Different architectures across vendors
- Inconsistent memory maps, boot flows and interrupt models
- Unpredictable timing from shared resources
- Rising safety and validation requirements
- Software rarely survives a hardware generation

SYSTEM CONTRACT	Vendor A REFERENCE	Vendor B	Vendor C
MEMORY MAP	0x8000_0000 base	0x2000_0000 base	vendor-defined
BOOT FLOW	ROM → SBI → payload	ROM → 2-stage BL	custom loader
INTERRUPT MODEL	CLINT + PLIC	CLIC	proprietary controller
TIMING BEHAVIOUR	shared L2, no isolation	partitioned	undocumented
RESULT	No shared system contract — every port, every validation campaign starts from zero.		

IMPACT

- Longer development cycles
- Higher integration and validation costs
- Fragmented ecosystems

RT-Europa addresses system-level fragmentation — not just instruction-level compatibility.

Architecture Family Context

Three Compute Classes. One System Approach

RT-Europa is one member of a broader architecture framework.

REAL-TIME SYSTEMS

Predictability first

- Deterministic behaviour
- Safety-critical applications
- Automotive focus

EMBEDDED SYSTEMS

Efficiency first

- Resource-constrained
- Power-efficient
- IoT and edge computing

APPLICATION-CLASS SYSTEMS

Throughput first

- High-performance
- Complex workloads
- AI and compute-intensive

SHARED ACROSS ALL THREE

The system contract

- System boot & init contract
- Memory map conventions
- Interrupt model
- Validation & trace evidence

Together they provide a consistent system approach across compute domains — so validation carries over instead of restarting.

Value Proposition We Specify Three Layers...

...and Prove the Bottom One

LAYERS 01—02

We define the contract

01 **PROFILE** SPECIFICATION

What the core must implement

ISA, extensions and privilege modes — selected with commercial focus, not only technical preference.

02 **PLATFORM** SPECIFICATION

What the system must guarantee

System-level SoC requirements for the target processor classes and markets.

LAYER 03

We prove it holds

03 **MICROARCHITECTURE** VALIDATION & CERTIFICATION

What is proven in silicon

Validation and certification of commercial solutions — evidence, not intent.

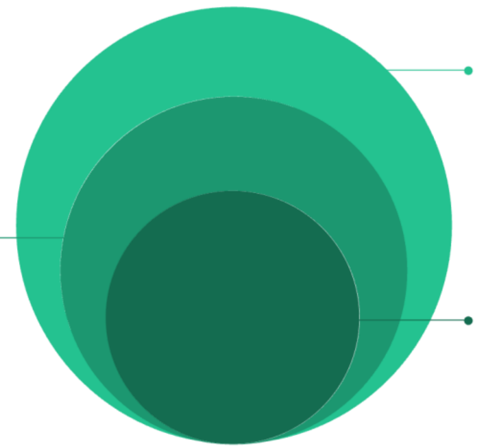
Specification and evidence come from the same source — that is what Professional Services delivers.

Profiles and Platforms: The Architectural Framework

Stabilizing System Behavior Across Vendors

- **Profiles = CPU baseline** curated ISA subsets plus privilege, exception and timing rules. Guarantees code portability at the instruction level.
- **Platforms = Subsystem baseline** memory map, boot/reset, debug, power and error reporting. Guarantees binary-level and tooling portability.
- **Why both matter:** profiles remove ISA fragmentation; platforms remove board-level fragmentation. Together they close the silicon-to-software gap for safety-critical use cases.

- **Current leadership:** chairing the RVM microcontroller profile, authoring the RT-Europa reference platform, coordinating upcoming embedded and application profiles



What is a RISC-V Platform?

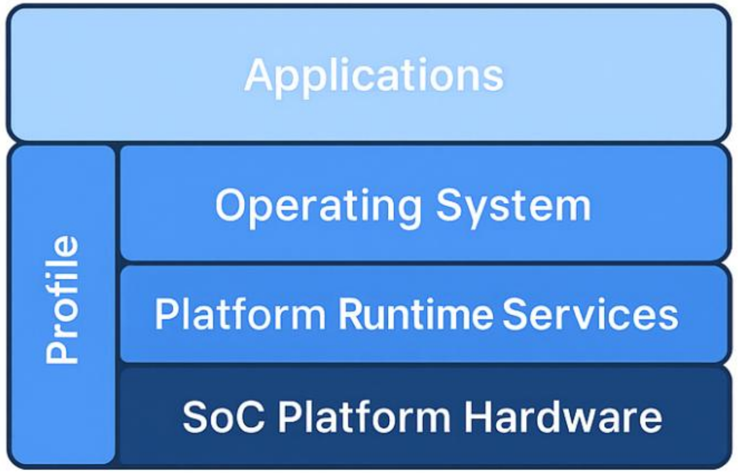
A RISC-V Platform defines the **firmware + hardware** environment required for running operating systems, ensuring **interoperability and consistency** across implementations.

Key Components

- **Profile** → Defines the base RISC-V ISA features and extensions.
- **Boot Runtime Services** → Functions needed during boot before OS handover.
- **Platform Runtime Services** → OS-level services (variables, time, virtual memory, reset).
- **SoC Platform Hardware** → Defines core SoC elements: CPUs, memory, peripherals, and integration rules.

Goal

Provide a **standardized environment** for OS and software developers to build portable, reliable RISC-V solutions.



Definitions – Extended Platform

The canonical definition of a RISC-V platform is not enough for one of the main goals of Quintauris: Reduce RISC-V defragmentation while avoiding lock-in and enabling RISC-V IP vendors competition in crucial differentiation features (i.e. pipeline).

There are definitions out of the RISC-V platform scope, that should be necessarily addressed or supported by Quintauris, as of:

- Compiler compatibility specifications→ ABI
- Debug interface compatibility specifications
- Performance
- Timing & Power constraints
- SoC Interconnect compatibility→ Bus interfaces
- Security requirements/add-ons (Cryptography extensions, modules or concepts –for example CHERI-)
- Safety requirements/add-ons (for example Safety mechanisms, Safety path signalling, etc.)

RT Europa Platform High Level Requirements

Deterministic Execution

- Predictable interrupt latency and bounded execution paths.
- Dedicated deterministic fabric for safety-critical tasks.
- Local and shared low-latency memory.

Scalable Performance

- Flexible configuration (1 to N cores, cache, CCM sizing, peripherals).
- Support for Floating Point and Vector Operations
- Support for Hypervisor and Device / Machine Virtualization

Standardized Software Interface

- RT-Europa Profile targets established RVI specs
- The profile is in process of being standardized (RVM TG)

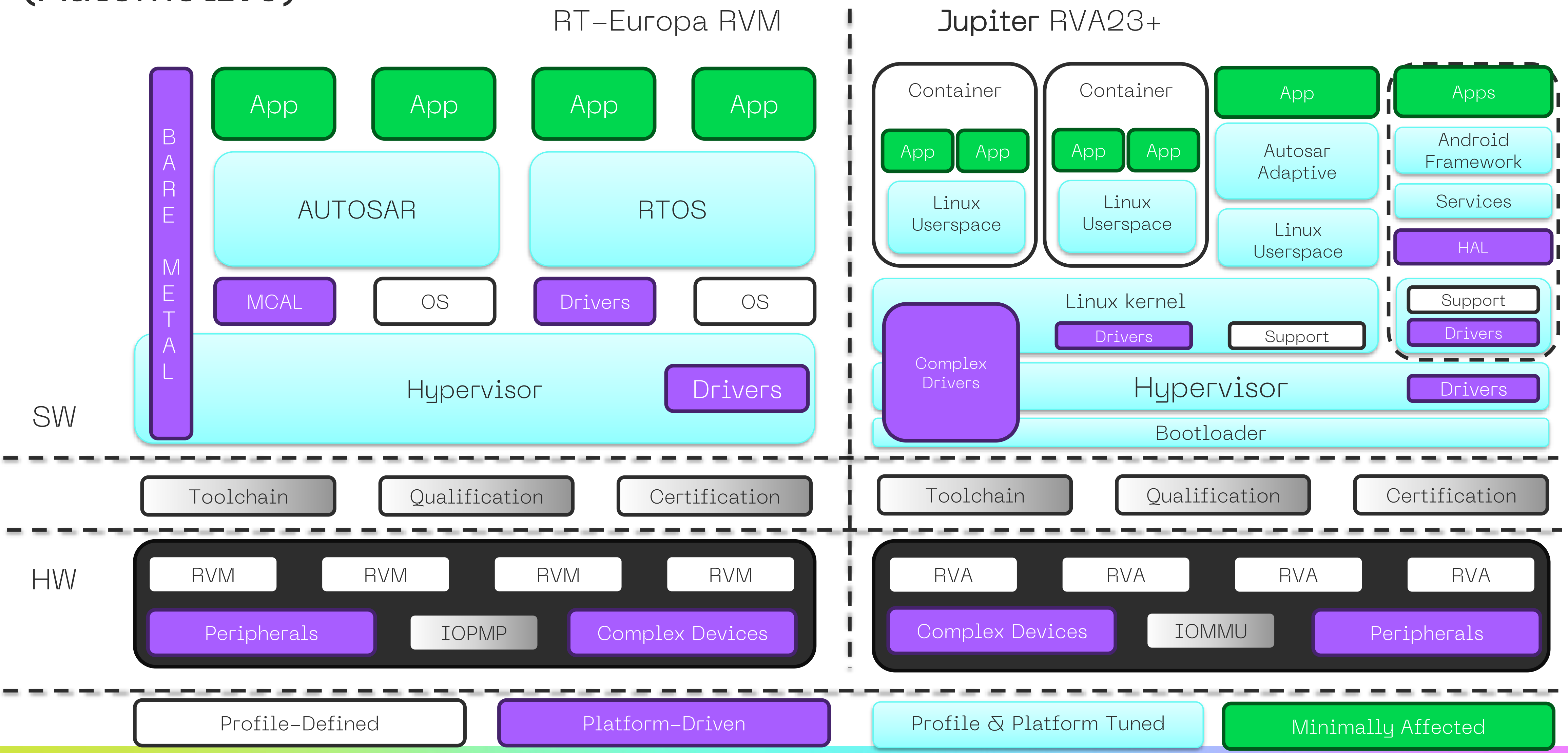
Robust Safety & Security Foundations

- ASIL-D capable, secure boot, and secure isolation.
- Explicit memory isolation (PMP/SPMP/IOPMP),
- Safety Mechanisms like Lock-Step, error detection (ECC), parity.

RT Europa Platform High Level Requirements

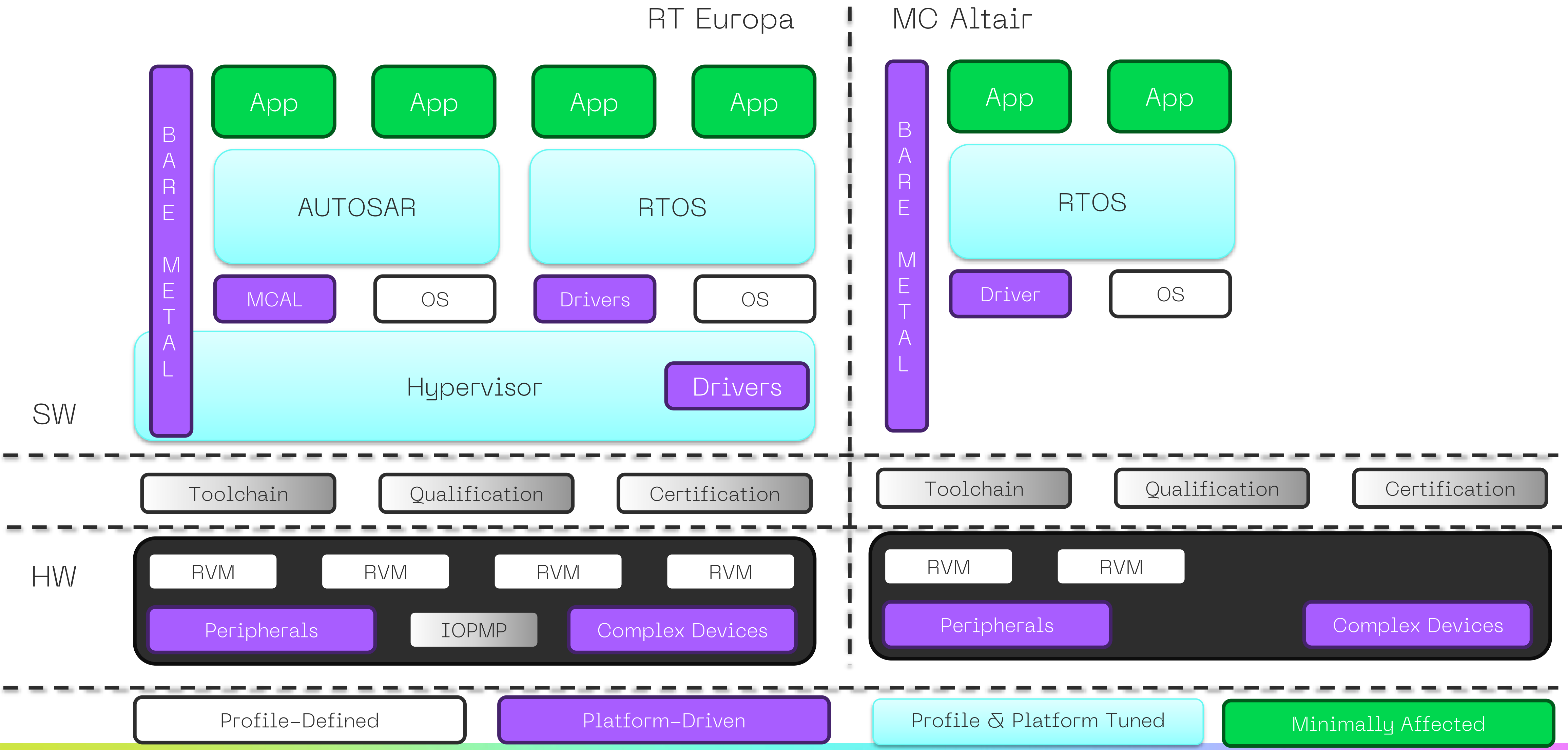
Flexible Software Integration	Explicit RTOS support (AUTOSAR, Zephyr, FreeRTOS, SafeRTOS). Compatible with bare-metal and hypervisor-based deployments.
Comprehensive Debug & Trace Infrastructure	Extensive debug & trace facilities for rapid bring-up and validation. Scalable debug support (single-core vs. multi-core domains).
Early Development Enablement	FPGA demonstration platforms for rapid prototyping. Virtual prototypes (SystemC/QEMU) for early SW development. Remote Device Farms for SW partners
Performance Benchmarking & Scalability	Clearly defined KPIs and benchmarking strategy. Scalable architecture to support diverse workloads efficiently.
Clear Certification & Extensibility Strategy	Pathway for certification (+fill the GAPs from RVI certification) Future-proofed design allowing variants (e.g., Altair derivatives).

Example Stack Composition Across RVM and RVA Definitions (Automotive)



Must involve some of these to gain traction: Not all SW companies join RVI

Example RVM Stack against Use Cases (Automotive)



Superset -> Subset makes it easier for SW vendors to manage their offerings

Agenda

Three Pillars of QNT Delivery

PILLAR 01

Benchmark & Evaluate

QNTAF-powered performance analysis

- Standardized benchmarks: CoreMark · Dhrystone · Embench IoT
- 11 core configurations · 6 OS targets · 4 toolchains
- Compilation profiles scoped per hardware configuration
- Insights dashboard · dual radar compare · PNG & CSV export

PILLAR 02

Validate & Certify

End-to-end traceability for automotive-grade confidence

- Design Packages → ACs → Xray → Robot → Argo CI
- Suites re-resolve from the matrix · campaigns launch a full sign-off
- Immutable trace snapshot per run — ISO 26262-aligned
- Automated pre-merge and pre-release CI gates

PILLAR 03

Deploy & Integrate

Hosted at infra.quintauris.com

- Kubernetes-native: Kind + ctlptl · Helm · Tilt · ArgoCD
- Placido launches a K8s Job per run · results on a shared PVC
- Fail-closed tenancy: platform registry + UI-feature entitlements
- Entra ID B2B guest SSO — first external customer live in v4.2

Measure Every Core. Compare Every Platform

- Run CoreMark, Dhrystone and Embench IoT 1.0 (size & speed) across RISC-V targets in a standardized, reproducible environment
- Simulation and FPGA hardware-in-the-loop: QEMU · Synopsys Virtualizer · Digilent Genesys 2 · AMD/Xilinx VCU118
- 11 core configurations across three processor classes — RealTime, Embedded, Application — including OpenHW CVA6, the RT-Europa family, Nuclei N900, Hazard3 and VexRiscV
- 7 OS targets: Bare Metal · FreeRTOS · SafeRTOS · ThreadX · Zephyr · EasyXMen · EmbOS
- Compilation profiles scoped per hardware configuration — toolchain, compiler and linker flags versioned with the test
- Results parsed by the Analytics Engine into the Insights dashboard — cross-platform compare, dual radar, PNG & CSV export

11
CORE CONFIGURATIONS

6
OS TARGETS BARE METAL · RTOS · AUTOSAR

4
COMPILER TOOLCHAINS GCC · CLANG/LLVM · NUCLEI

3
BENCHMARK SUITES COREMARK · DHRYSTONE · EMBENCH

Traceability from Requirement to Evidence

01
Design Packages
Jira / QNTAF

02
Acceptance Criteria
BDD Format

03
Xray Tests & VVC Sets
Xray / VVC

04
Robot Framework Automation
Robot / Python

05
Argo Workflows CI Gate
Argo / Xray

BDD-DRIVEN DESIGN

Every requirement expressed as executable acceptance criteria

Design Packages authored in Jira, criteria in BDD format, linked to Xray test sets

SUITES & CAMPAIGNS

A full release sign-off in a single launch

Suites re-resolve from the compatibility matrix; a campaign runs many suites as one nightly or release gate

IMMUTABLE EVIDENCE

Every run sealed with a trace snapshot at dispatch

ISO 26262-aligned trace metadata plus an append-only audit log — company-scoped and tamper-evident

Production-Grade Platform

Hosted at infra.quintauris.com

01 QNTAF Application	Frontend (React 19 / Vite 7) · Backend (Flask + SQLAlchemy) · Placido (FastAPI) · PostgreSQL 15
02 Container Platform	Kubernetes · Kind + ctlptl · Helm chart · Tilt · ingress-nginx
03 Infra Services	ArgoCD · Argo Workflows · Harbor · Nexus · SonarQube · Vault · Grafana · Prometheus
04 KVM Cluster	22 vCPU · ~197 GB RAM · ~1 TB storage · dev / staging / production
05 Hardware Lab	FPGA lab nodes · SoC targets · JTAG debug probes · power controllers

EXECUTION MODEL

Placido launches one Kubernetes Job per run · results written to a shared PVC · logs stream Job → Placido → backend → browser in real time

TENANCY & ACCESS

Fail-closed per-company platform registry and UI-feature entitlements · Entra ID B2B guest SSO · durable PostgreSQL migrated in place by Alembic

LIVE DEMO

INFRA: DevOps Infrastructure Backbone

INFRA is QNT's infrastructure-as-code backbone: it **defines and deploys the full DevOps environment** that hosts and supports our internal tools and services — including QNTAF.

Automation-first	Reproducible	Version-controlled	Self-service

COMPONENTS

infra	Defines and deploys the full DevOps infrastructure as code — the single source of truth for the platform.
infra-virt	Automated Ubuntu VM setup via Multipass + Ansible: a ready-to-use DevOps environment in minutes.
Platform docs	Living documentation of the platform and its services (infra.quintauris.com).

One Backbone Behind Every Stage

01 · UNIFIED

Unified infrastructure for all stages

Simulation, FPGA hardware-in-the-loop and native RISC-V execution — all run in the cloud from one environment.

02 · PORTABLE

Cloud-native & locally deployable

A Docker-based design means the same stack deploys for internal teams or inside an external partner's environment.

03 · AUTOMATED

Integrated into CI/CD workflows

Compatible with Jenkins and GitHub for continuous validation, benchmarking and regression checks.

04 · TRACEABLE

Full traceability by design

Artifacts, configurations and results are logged automatically — evidence ready for audits and certification.

05 · SCALABLE

Scales across the company

One shared platform serves multiple teams and projects: SoC design, software validation, external IP repository integration.

PRACTICAL USE CASE

SoC vendor deploys QNTAF remotely via Docker

Tests run in their own cloud-hosted pipelines, with full trace capture and compliance reporting flowing back.

Agenda

- 01 QNT Specifications
- 02 Infrastructure
- 03 Test Automation Framework
- 04 OS Integrations
- 05 Wrap-Up & Discussion

LIVE DEMO

v4.2.0

QNT Automation Test Framework

Web-based RISC-V benchmarking & test-automation platform. Runs standardized benchmarks — CoreMark, Dhrystone, embench IoT — across RISC-V processors on **FPGA hardware and simulators.**

CORE CAPABILITIES

- Test orchestration across heterogeneous RISC-V environments
- Automated benchmark execution, logging & reporting
- Containerized, reproducible toolchains
- Data capture, analysis & web dashboards
- Web portal for selection, control & result browsing
- Plugin extensibility for new targets, simulators & tools

ARCHITECTURE

- **Backend** Flask / Python: JWT auth, SQLAlchemy, Socket.IO
  Frontend React / Vite: Charts visualization
- **PD** Orchestrator
- **Database** PostgreSQL
- **E2E tests** Robot Framework
- **Runtime** Minikube + Tilt (Kubernetes)

Governance.

- Role-based access (viewer < engineer < admin)
- Usage tiers (evaluation < validation < certification)
- Full audit logging

QNTAF — Test Automation Framework

How it works / Core Platform Capabilities

Core Platform Capabilities

Test Orchestration

Manage and execute functional, performance, and real-time tests across heterogeneous RISC-V environments from a single control plane.

Benchmark Execution

Run industry-standard benchmarks (CoreMark, Dhrystone) with accurate timing, logging, and reproducible reporting across physical and virtual targets.

RBAC & Audit Trail

Role-based access control (viewer → engineer → admin) with full audit logging and usage tier controls (evaluation → validation → certification).

Real-Time Log Streaming

Room-based WebSocket architecture delivers targeted live execution logs to each user session without cross-contamination between test runs.

Plugin Architecture

Extensible analysis engine supports new test types, hardware targets, simulators, and benchmarking tools without core rewrites.

Certification Evidence

ISO-aligned PDF evidence endpoint links benchmark results, trace metadata snapshots, and audit excerpts — the first step from benchmark runner to certification platform.

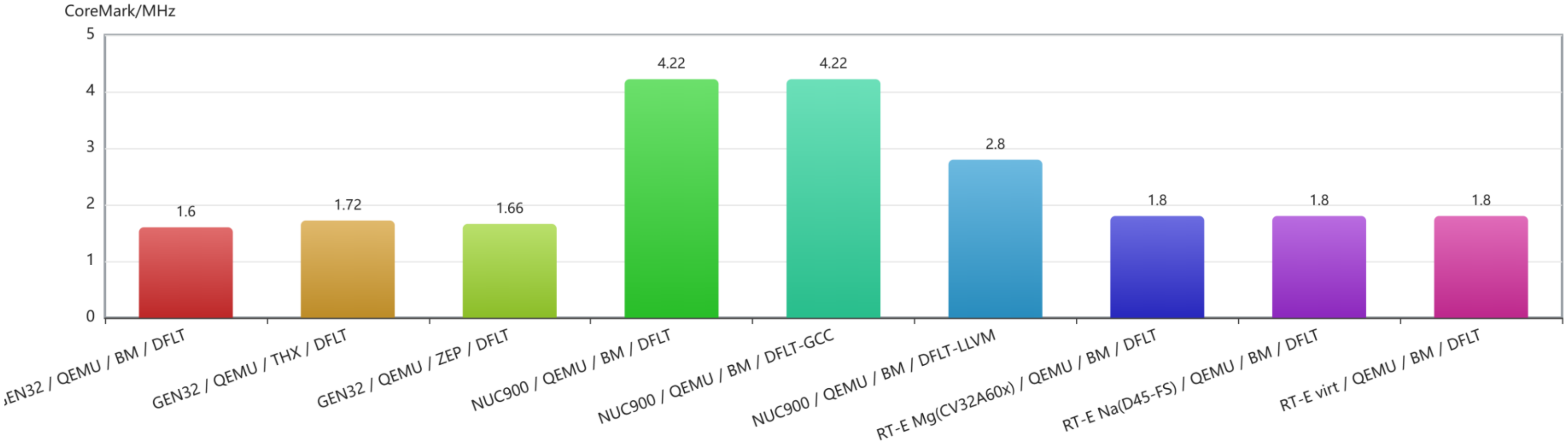
QNTAF — Test Automation Framework

Capability Matrix

		ENVIRONMENT	CORE IP	OPERATING SYSTEM	COMPILER	DEBUGGER
VIRTUAL	SIMULATION	Synopsys Virtualizer	SYNOPSYS RHX110FS	SafeRTOS, iSoft, FreeRTOS, Bare Metal	GCC, CLANG	SEGGER, LAUTERBACH, TASKING
		QEMU	HAZARD 3, CVA6, RT-EUROPA, NUCLEI N900, ANDES D45 SE	Bare Metal, ThreadX, Zephyr	GCC, CLANG	GDB
PHYSICAL	FPGA	Genesys 2	CVA6, ANDES D45 SE	FreeRTOS, SafeRTOS, ThreadX, Bare Metal	iAR, HighTec, CLANG, GCC	SEGGER, LAUTERBACH
		Xilinx XCU	SYNOPSYS RHX110FS	SafeRTOS	Tasking	SEGGER, LAUTERBACH, Tasking, GDB
		Arty A7	HAZARD 3	Zephyr	GCC	GDB
	SOC	HPM6200	ANDES D45 SE	Vector Microsar	Tasking	SEGGER, LAUTERBACH
		HPM6750	ANDES D45 SE	ThreadX	Tasking	SEGGER, LAUTERBACH
		Raspberry Pi Pico 2	HAZARD 3	Zephyr	GCC	GDB

Results comparison 1/3

Single-metric ranking



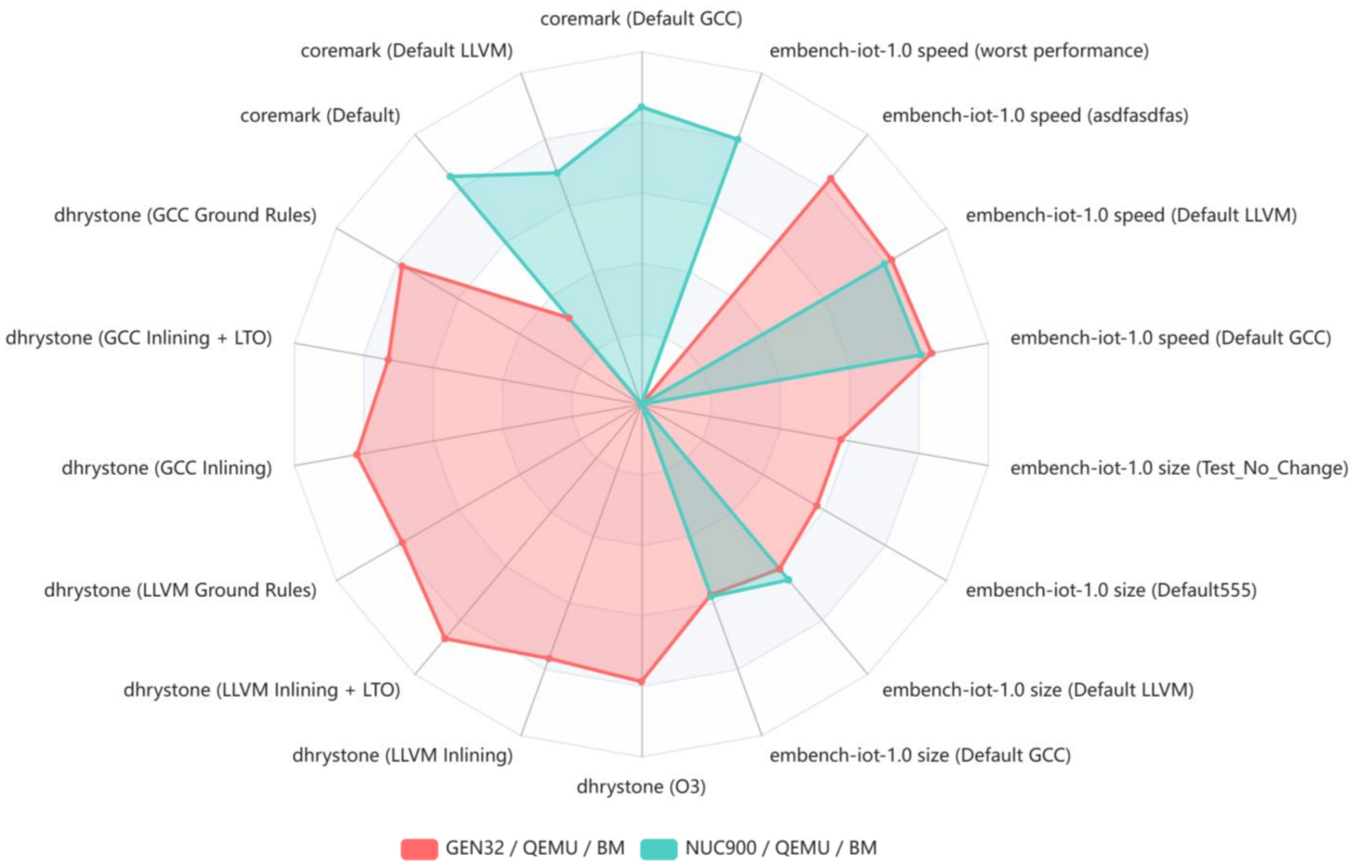
Single normalized metric (CoreMark/MHz) across configurations, quick side-by-side ranking:

- Reduces each configuration to one comparable score
- Ranks configurations side by side at a glance
- Complements the deeper per-benchmark views

Disclaimer: values shown are mocked sample data for illustration only, not real performance.

Results comparison 1/3

Multi-suite platform comparison



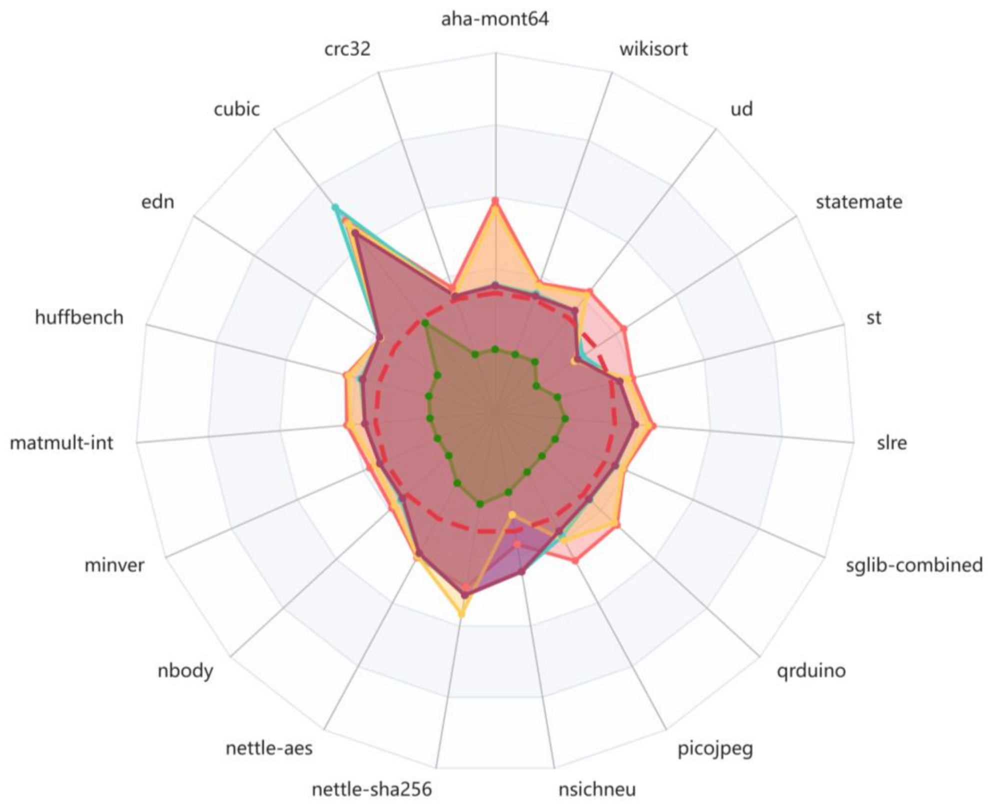
Contrast configurations across CoreMark, Dhrystone, and Embench axes at once.
It makes it easy to see how relative strengths shift by workload type:

- Overlays multiple benchmark suites on a single view
- Shows how relative strengths shift by workload type
- Avoids collapsing the comparison to a single winner

Disclaimer: values shown are mocked sample data for illustration only, not real performance.

Results comparison 3/3

Embench IoT per-benchmark comparison



Ability to break performance down across individual Embench IoT workloads, overlaying multiple configurations so per-benchmark strengths and weaknesses are visible at a glance against a reference baseline.

- Breaks results down to individual Embench IoT workloads
- Overlays multiple configurations on one chart
- Highlights per-benchmark strengths and weaknesses against a reference baseline

Disclaimer: values shown are mocked sample data for illustration only, not real performance.

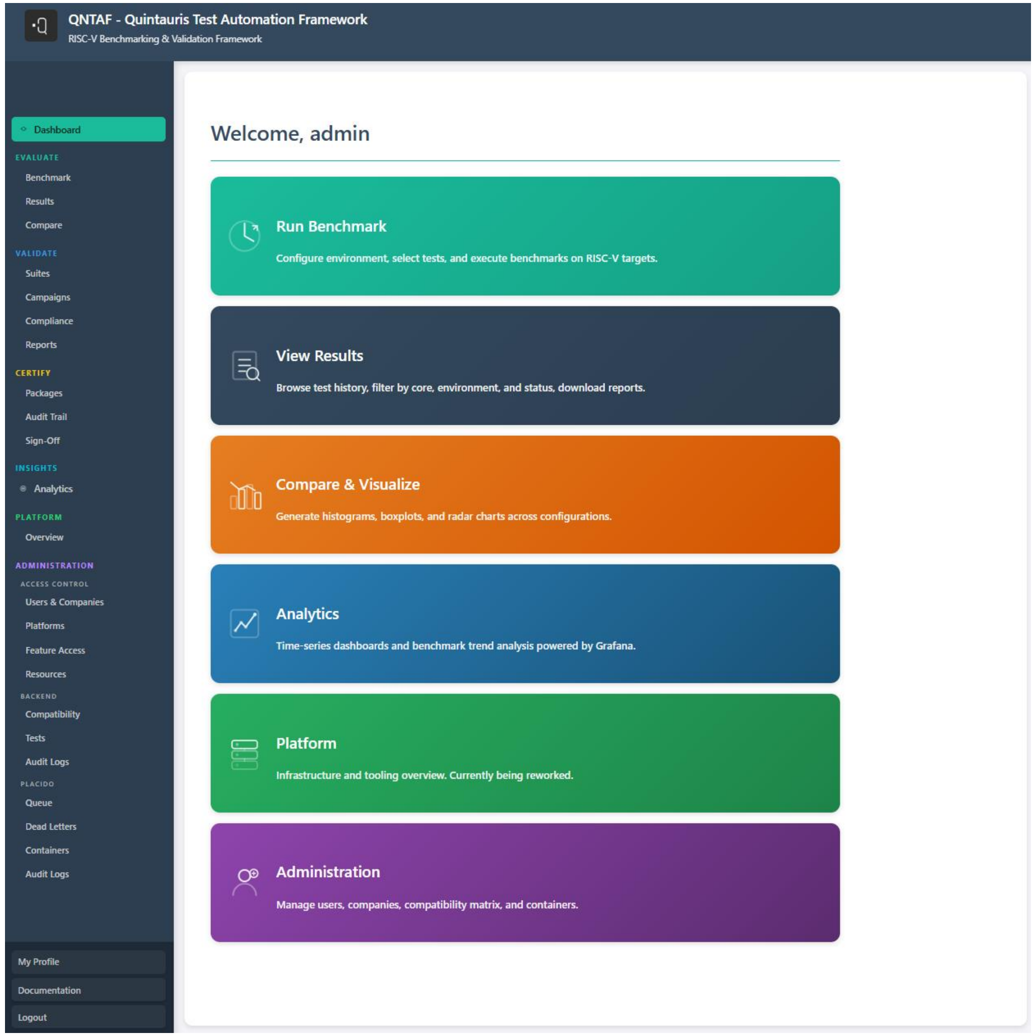
QNTAF Workflows

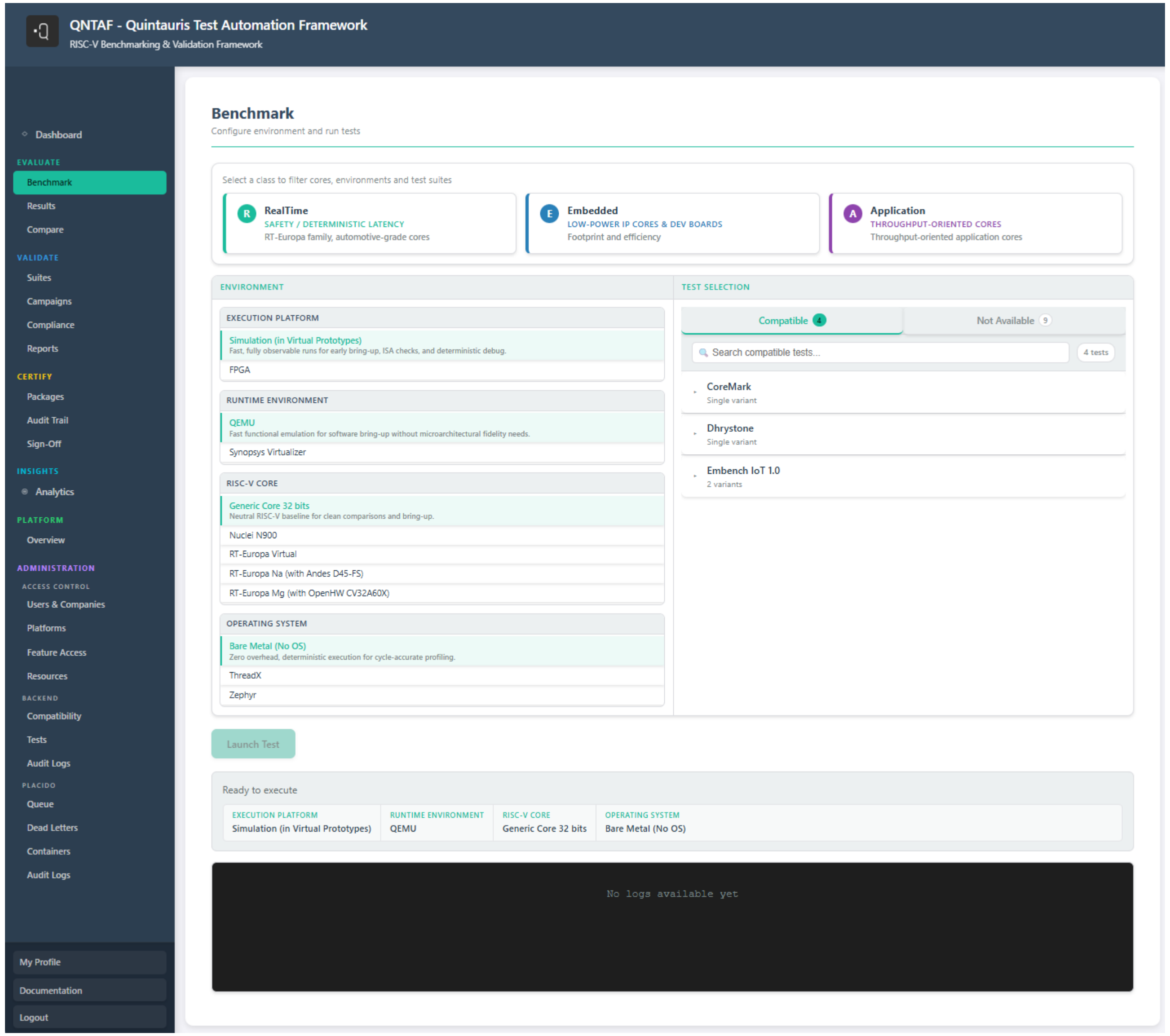
1. **Benchmarking: RISC-V IP Comparison in Hybrid Environments**
   Use reference platforms for consistent IP performance benchmarking across simulation, cloud, and physical targets to support adoption decisions.

2. **Validation and Certification**
   Deploy platforms as initial certification targets to validate compliance with defined profiles and requirements on real systems.

Coffee Break

Demo – QNTAF


QNTAF - Quintauris Test Automation Framework
RISC-V Benchmarking & Validation Framework
Dashboard
EVALUATE
Benchmark
Results
Compare
VALIDATE
Suites
Campaigns
Compliance
Reports
CERTIFY
Packages
Audit Trail
Sign-Off
INSIGHTS
Analytics
PLATFORM
Overview
ADMINISTRATION
ACCESS CONTROL
Users & Companies
Platforms
Feature Access
Resources
BACKEND
Compatibility
Tests
Audit Logs
PLACIDO
Queue
Dead Letters
Containers
Audit Logs
My Profile
Documentation
Logout
Welcome, admin
Run Benchmark
Configure environment, select tests, and execute benchmarks on RISC-V targets.
View Results
Browse test history, filter by core, environment, and status, download reports.
Compare & Visualize
Generate histograms, boxplots, and radar charts across configurations.
Analytics
Time-series dashboards and benchmark trend analysis powered by Grafana.
Platform
Infrastructure and tooling overview. Currently being reworked.
Administration
Manage users, companies, compatibility matrix, and containers.


QNTAF - Quintauris Test Automation Framework
RISC-V Benchmarking & Validation Framework
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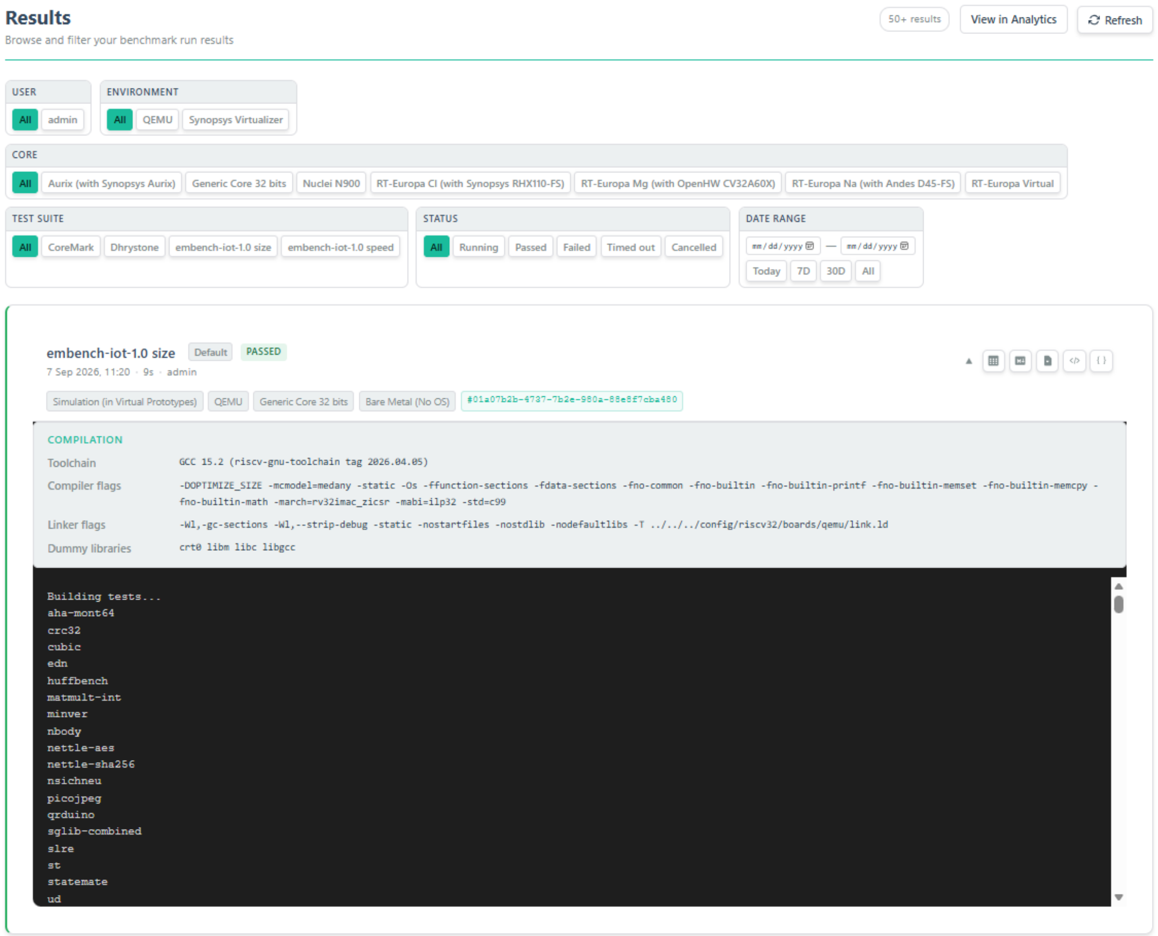
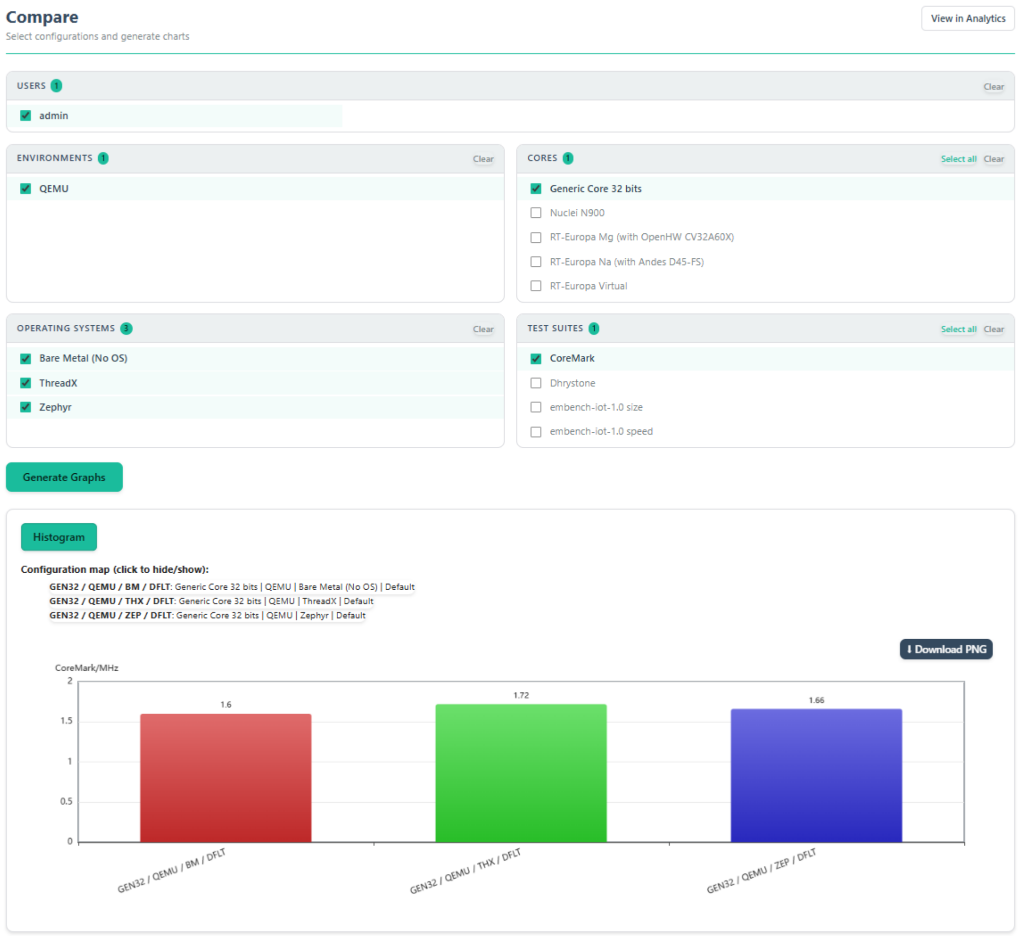
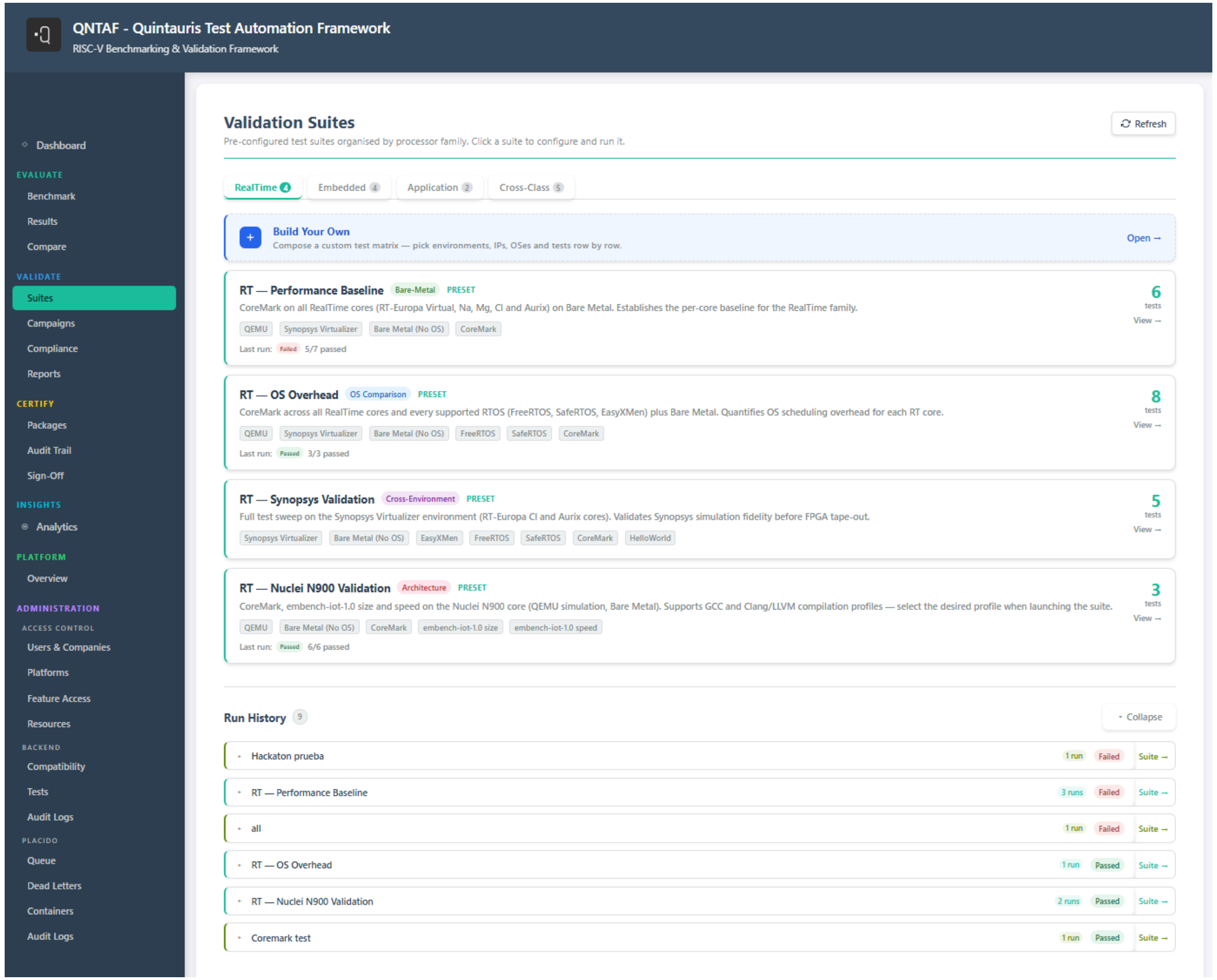
Campaigns
Compliance
Reports
CERTIFY
Packages
Audit Trail
Sign-Off
INSIGHTS
Analytics
PLATFORM
Overview
ADMINISTRATION
ACCESS CONTROL
Users & Companies
Platforms
Feature Access
Resources
BACKEND
Compatibility
Tests
Audit Logs
PLACIDO
Queue
Dead Letters
Containers
Audit Logs
My Profile
Documentation
Logout
Benchmark
Configure environment and run tests
Select a class to filter cores, environments and test suites
RealTime
SAFETY / DETERMINISTIC LATENCY
RT-Europa family, automotive-grade cores
Embedded
LOW-POWER IP CORES & DEV BOARDS
Footprint and efficiency
Application
THROUGHPUT-ORIENTED CORES
Throughput-oriented application cores
ENVIRONMENT
EXECUTION PLATFORM
Simulation (in Virtual Prototypes)
Fast, fully observable runs for early bring-up, ISA checks, and deterministic debug.
FPGA
RUNTIME ENVIRONMENT
QEMU
Fast functional emulation for software bring-up without microarchitectural fidelity needs.
Synopsys Virtualizer
RISC-V CORE
Generic Core 32 bits
Neutral RISC-V baseline for clean comparisons and bring-up.
Nuclei N900
RT-Europa Virtual
RT-Europa Na (with Andes D45-FS)
RT-Europa Mg (with OpenHW CV32A60X)
OPERATING SYSTEM
Bare Metal (No OS)
Zero overhead, deterministic execution for cycle-accurate profiling.
ThreadX
Zephyr
TEST SELECTION
Compatible 4
Not Available 9
Search compatible tests...
4 tests
CoreMark
Single variant
Dhrystone
Single variant
Embench IoT 1.0
2 variants
Launch Test
Ready to execute
EXECUTION PLATFORM
Simulation (in Virtual Prototypes)
RUNTIME ENVIRONMENT
QEMU
RISC-V CORE
Generic Core 32 bits
OPERATING SYSTEM
Bare Metal (No OS)
No logs available yet

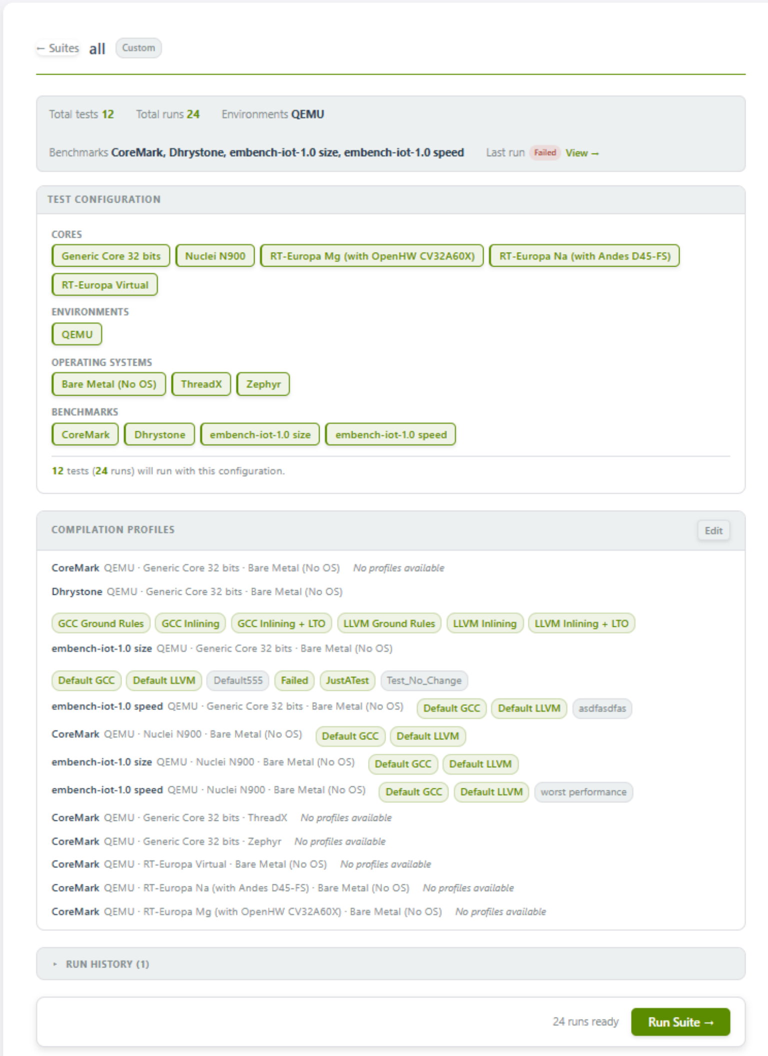
1. Evaluation



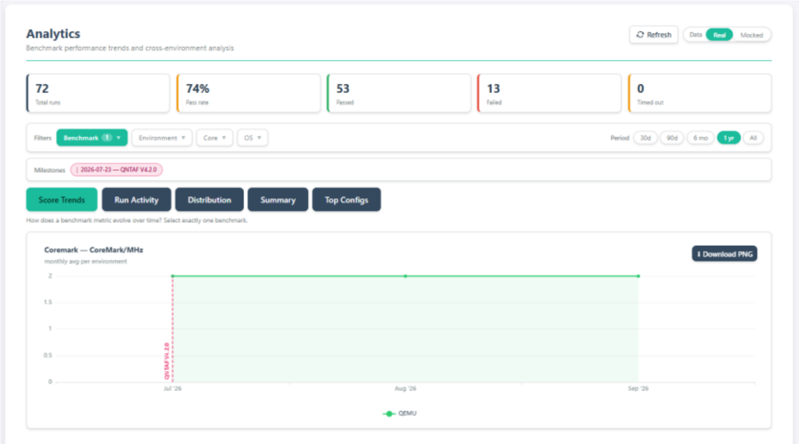


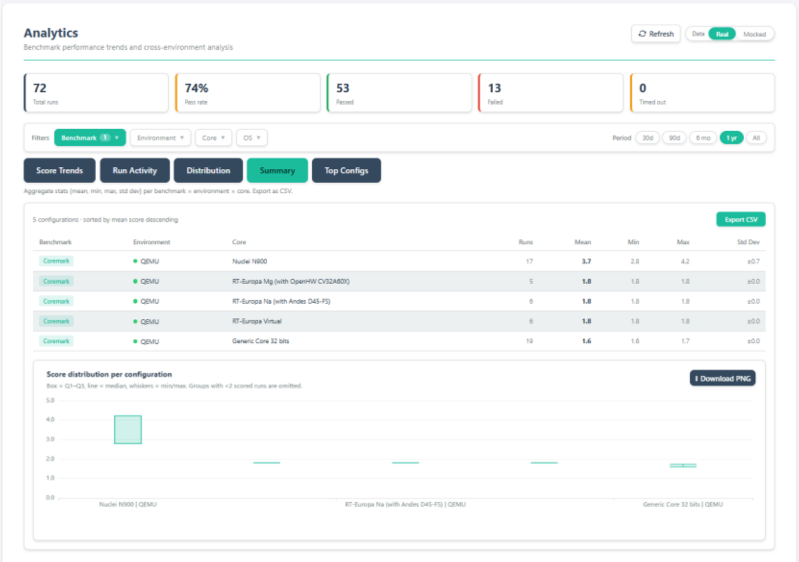
2. Validation

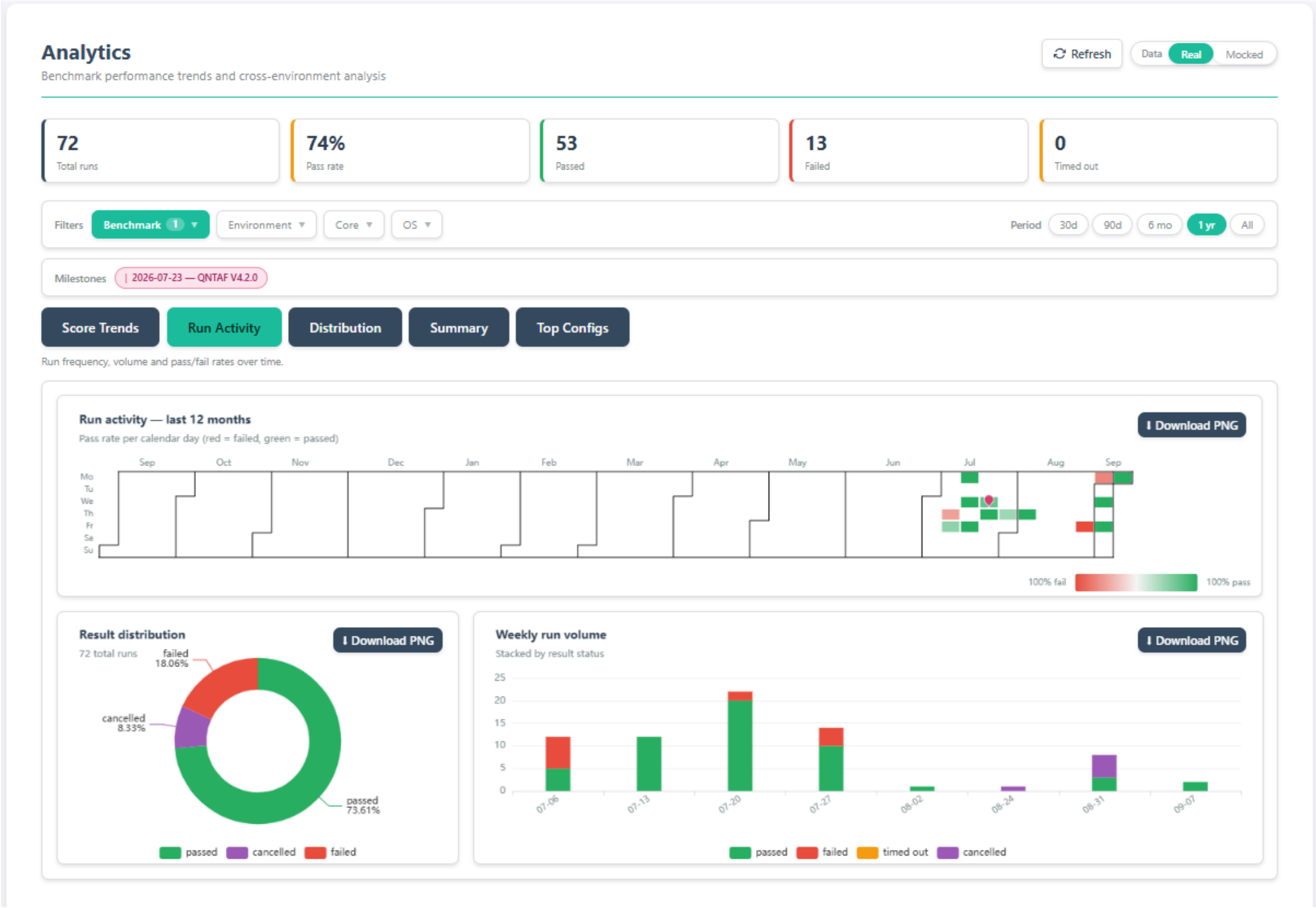




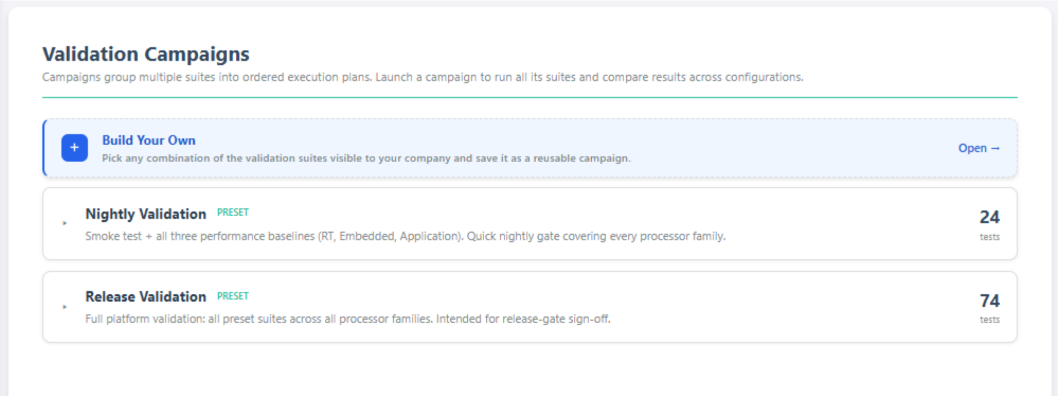
Comparison and analytics







Campaigns, Compliance and Reports



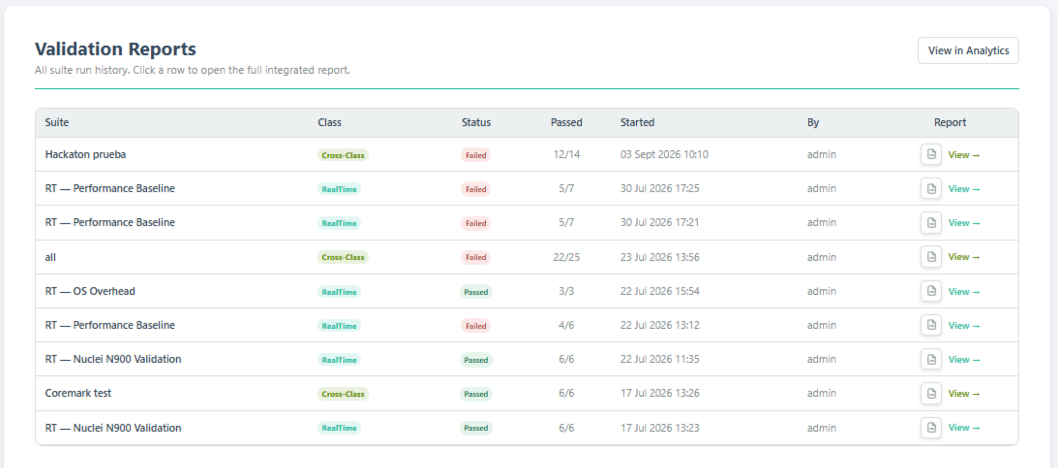
Validation Campaigns

Campaigns group multiple suites into ordered execution plans. Launch a campaign to run all its suites and compare results across configurations.

Build Your Own
Pick any combination of the validation suites visible to your company and save it as a reusable campaign.
Open →

Nightly Validation PRESET
Smoke test + all three performance baselines (RT, Embedded, Application). Quick nightly gate covering every processor family.
24 tests

Release Validation PRESET
Full platform validation: all preset suites across all processor families. Intended for release-gate sign-off.
74 tests

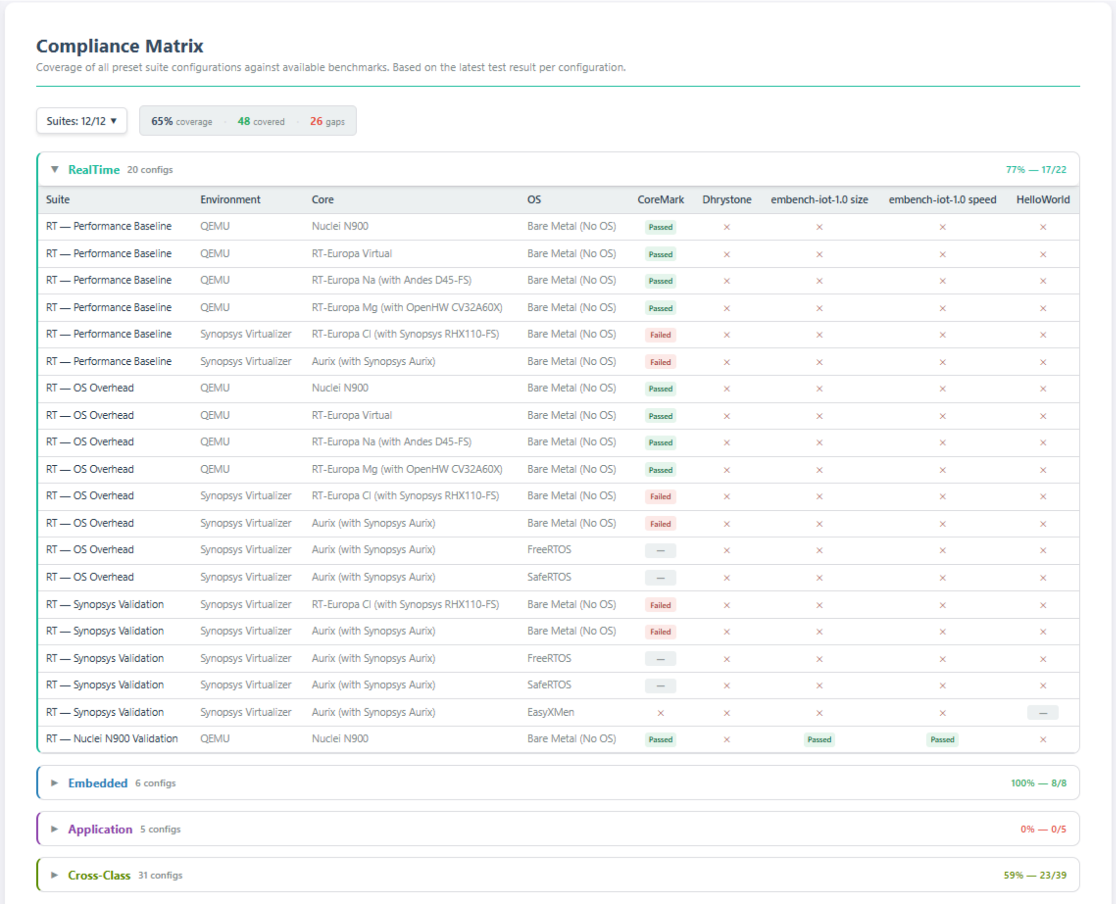


Validation Reports

View in Analytics

All suite run history. Click a row to open the full integrated report.

Suite	Class	Status	Passed	Started	By	Report
Hackaton prueba	Cross-Class	Failed	12/14	03 Sept 2026 10:10	admin	View →
RT — Performance Baseline	RealTime	Failed	5/7	30 Jul 2026 17:25	admin	View →
RT — Performance Baseline	RealTime	Failed	5/7	30 Jul 2026 17:21	admin	View →
all	Cross-Class	Failed	22/25	23 Jul 2026 13:56	admin	View →
RT — OS Overhead	RealTime	Passed	3/3	22 Jul 2026 15:54	admin	View →
RT — Performance Baseline	RealTime	Failed	4/6	22 Jul 2026 13:12	admin	View →
RT — Nuclei N900 Validation	RealTime	Passed	6/6	22 Jul 2026 11:35	admin	View →
Coremark test	Cross-Class	Passed	6/6	17 Jul 2026 13:26	admin	View →
RT — Nuclei N900 Validation	RealTime	Passed	6/6	17 Jul 2026 13:23	admin	View →



Compliance Matrix

Coverage of all preset suite configurations against available benchmarks. Based on the latest test result per configuration.

Suites: 12/12 ▾ | 65% coverage · 48 covered · 26 gaps

▾ **RealTime** 20 configs — 77% — 17/22

Suite	Environment	Core	OS	CoreMark	Dhrystone	embench-iot-1.0 size	embench-iot-1.0 speed	HelloWorld
RT — Performance Baseline	QEMU	Nuclei N900	Bare Metal (No OS)	Passed	×	×	×	×
RT — Performance Baseline	QEMU	RT-Europa Virtual	Bare Metal (No OS)	Passed	×	×	×	×
RT — Performance Baseline	QEMU	RT-Europa Na (with Andes D45-FS)	Bare Metal (No OS)	Passed	×	×	×	×
RT — Performance Baseline	QEMU	RT-Europa Mg (with OpenHW CV32A60X)	Bare Metal (No OS)	Passed	×	×	×	×
RT — Performance Baseline	Synopsys Virtualizer	RT-Europa Cl (with Synopsys RHX110-FS)	Bare Metal (No OS)	Failed	×	×	×	×
RT — Performance Baseline	Synopsys Virtualizer	Aurix (with Synopsys Aurix)	Bare Metal (No OS)	Failed	×	×	×	×
RT — OS Overhead	QEMU	Nuclei N900	Bare Metal (No OS)	Passed	×	×	×	×
RT — OS Overhead	QEMU	RT-Europa Virtual	Bare Metal (No OS)	Passed	×	×	×	×
RT — OS Overhead	QEMU	RT-Europa Na (with Andes D45-FS)	Bare Metal (No OS)	Passed	×	×	×	×
RT — OS Overhead	QEMU	RT-Europa Mg (with OpenHW CV32A60X)	Bare Metal (No OS)	Passed	×	×	×	×
RT — OS Overhead	Synopsys Virtualizer	RT-Europa Cl (with Synopsys RHX110-FS)	Bare Metal (No OS)	Failed	×	×	×	×
RT — OS Overhead	Synopsys Virtualizer	Aurix (with Synopsys Aurix)	Bare Metal (No OS)	Failed	×	×	×	×
RT — OS Overhead	Synopsys Virtualizer	Aurix (with Synopsys Aurix)	FreeRTOS	—	×	×	×	×
RT — OS Overhead	Synopsys Virtualizer	Aurix (with Synopsys Aurix)	SafeRTOS	—	×	×	×	×
RT — Synopsys Validation	Synopsys Virtualizer	RT-Europa Cl (with Synopsys RHX110-FS)	Bare Metal (No OS)	Failed	×	×	×	×
RT — Synopsys Validation	Synopsys Virtualizer	Aurix (with Synopsys Aurix)	Bare Metal (No OS)	Failed	×	×	×	×
RT — Synopsys Validation	Synopsys Virtualizer	Aurix (with Synopsys Aurix)	FreeRTOS	—	×	×	×	×
RT — Synopsys Validation	Synopsys Virtualizer	Aurix (with Synopsys Aurix)	SafeRTOS	—	×	×	×	×
RT — Synopsys Validation	Synopsys Virtualizer	Aurix (with Synopsys Aurix)	EasyXMen	×	×	×	×	—
RT — Nuclei N900 Validation	QEMU	Nuclei N900	Bare Metal (No OS)	Passed	×	Passed	Passed	×

▸ **Embedded** 6 configs — 100% — 8/8

▸ **Application** 5 configs — 0% — 0/5

▸ **Cross-Class** 31 configs — 59% — 23/39

3. Certification (on the roadmap)

Certification Packages

Build formal certification packages that bundle test evidence, compliance matrices, and validation reports into auditable deliverables aligned with industry standards.

+ New Package

Package	Standard	Created	Completeness	Status
No certification packages created yet.				

Audit Trail

Complete, immutable record of all actions taken during the certification process. Every test execution, report generation, sign-off, and configuration change is logged.

Timestamp	User	Action	Resource	Details
No audit events recorded yet.				

Digital Sign-Off

Multi-stage approval workflow for certification packages. Authorized signatories review and digitally approve each stage before a package can be released.

Package	Stage	Assignee	Status	Signed At
No sign-off requests pending.				

Agenda

01 QNT Specifications

02 Infrastructure

03 Test Automation Framework

04 OS Integrations

05 Wrap-Up & Discussion

QNTAF — Test Automation Framework

Capability Matrix

		ENVIRONMENT	CORE IP	OPERATING SYSTEM	COMPILER	DEBUGGER
VIRTUAL	SIMULATION	Synopsys Virtualizer	SYNOPSYS RHX110FS	SafeRTOS, iSoft, FreeRTOS, Bare Metal	GCC, CLANG	SEGGER, LAUTERBACH, TASKING
		QEMU	HAZARD 3, CVA6, RT-EUROPA, NUCLEI N900, ANDES D45 SE	Bare Metal, ThreadX, Zephyr	GCC, CLANG	GDB
PHYSICAL	FPGA	Genesys 2	CVA6, ANDES D45 SE	FreeRTOS, SafeRTOS, ThreadX, Bare Metal	iAR, HighTec, CLANG, GCC	SEGGER, LAUTERBACH
		Xilinx XCU	SYNOPSYS RHX110FS	SafeRTOS	Tasking	SEGGER, LAUTERBACH, Tasking, GDB
		Arty A7	HAZARD 3	Zephyr	GCC	GDB
	SOC	HPM6200	ANDES D45 SE	Vector Microsar	Tasking	SEGGER, LAUTERBACH
		HPM6750	ANDES D45 SE	ThreadX	Tasking	SEGGER, LAUTERBACH
		Raspberry Pi Pico 2	HAZARD 3	Zephyr	GCC	GDB

2026 RT Europa
Three Demo Stacks –One Constant Base Layer

REF ARCHITECTURE 1
Safety RTOS baseline

Demo App SW

Safe RTOS

QNT Base Layer — QUINTAURIS

HW IP REFERENCE
Andes D45-SE

DEBUG	TOOLCHAIN
Lauterbach	Tasking

REF ARCHITECTURE 2
Same stack, different silicon

Demo App SW

Safe RTOS

QNT Base Layer — QUINTAURIS

HW IP REFERENCE
Synopsys RHX-110-FS

DEBUG	TOOLCHAIN
Lauterbach	Tasking

REF ARCHITECTURE 3
AUTOSAR setup · HPMicro

Demo App SW

Vector MICROSAR

QNT Base Layer — QUINTAURIS

HW IP REFERENCE
Andes D45-SE

DEBUG	TOOLCHAIN
Segger	IAR

Vary the silicon, vary the RTOS and the tools — the QNT Base Layer and the validation evidence stay the same. HW IP references are starting points.

2027 M-Altair

Three Markets. Three Tool Ecosystems. Same Base Layer

REF ARCHITECTURE 1	REF ARCHITECTURE 2	REF ARCHITECTURE 3
Industrial	Automotive	Open source
Demo App SW	Demo App SW	Demo App SW
EmbOS	Safe RTOS	Zephyr
QNT Base Layer (QUINTAURIS)	QNT Base Layer (QUINTAURIS)	QNT Base Layer (QUINTAURIS)
HW IP REFERENCE: Andes N25F-SE	HW IP REFERENCE: MIPS RMX	HW IP REFERENCE: Hazard3
DEBUG: Segger	DEBUG: Tasking	DEBUG: OpenOCD
TOOLCHAIN: IAR	TOOLCHAIN: HighTec (LLVM)	TOOLCHAIN: GCC

Industrial, automotive and fully open source — the QNT Base Layer keeps all three validated by one flow. HW IP references are starting points.

Software Stack Model

- Bootloader and system startup
- Firmware initialization
- RTOS or bare-metal environments
- Optional operating systems or hypervisors
- Drivers and hardware discovery
- Safety and monitoring services

LAYERS 4—5

Software Responsibility

Application Logic 05

Drivers & Services 04

LAYERS 1—3

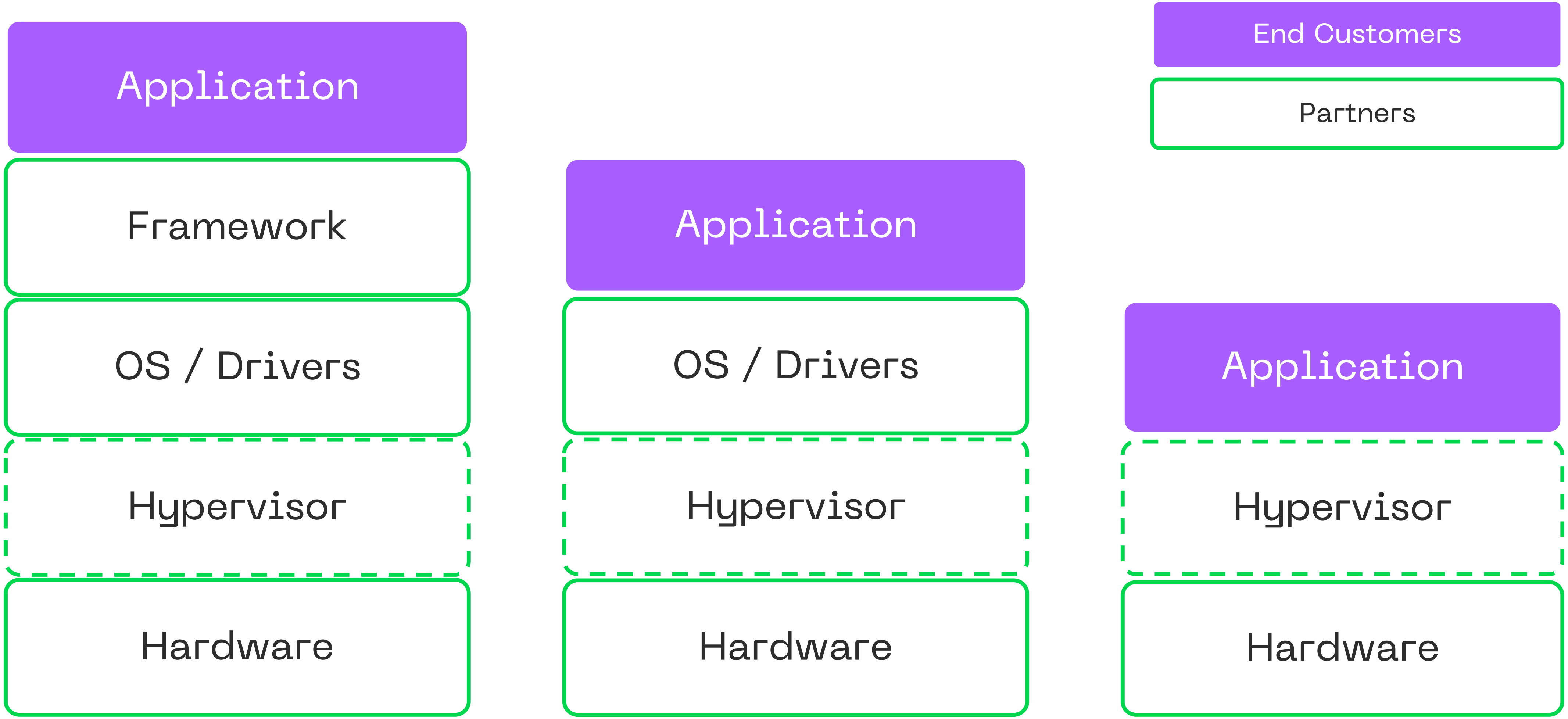
Platform Responsibility

RTOS / OS / Hypervisor 03

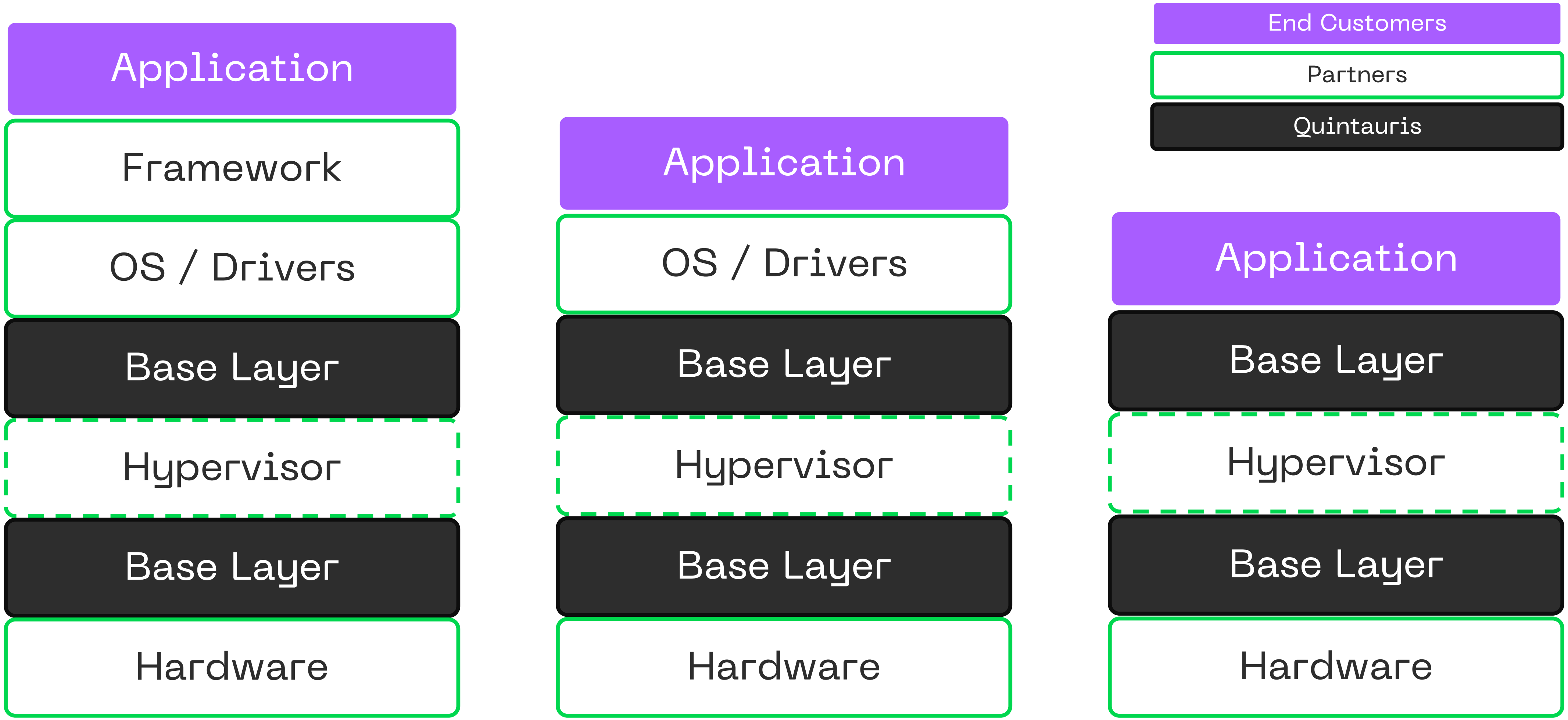
Bootloader & Firmware 02

Hardware Platform 01

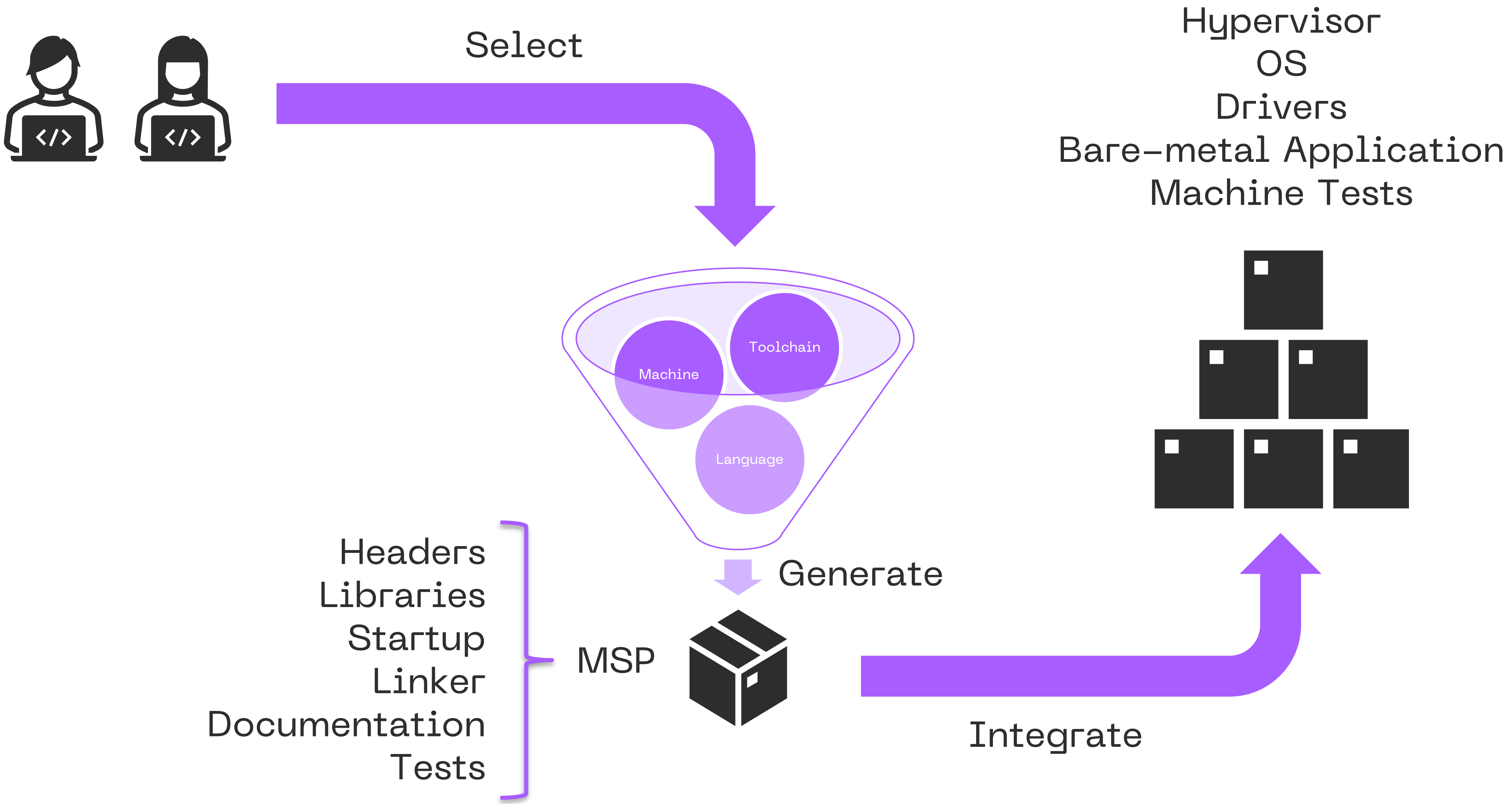
Real-Time Software Stacks



Real-Time Software Stacks



SoC Vendors | Software Base Layer



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